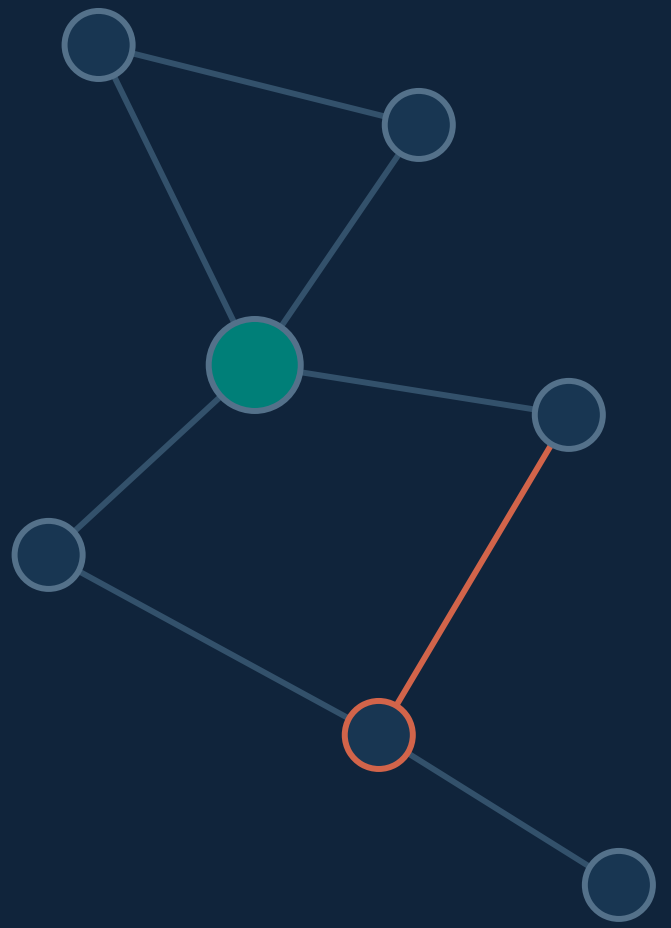
Beyond Weisfeiler–Lehman

Persistent and cycle-aware graph representations

Implement established methods. Modify cycle features. Measure the effect.

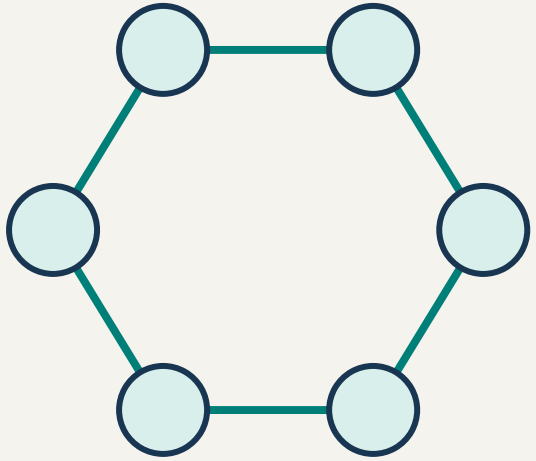
WL · P-WL · P-WL-C · λ weighting · Binary cycles



Graph structure carries more than node counts

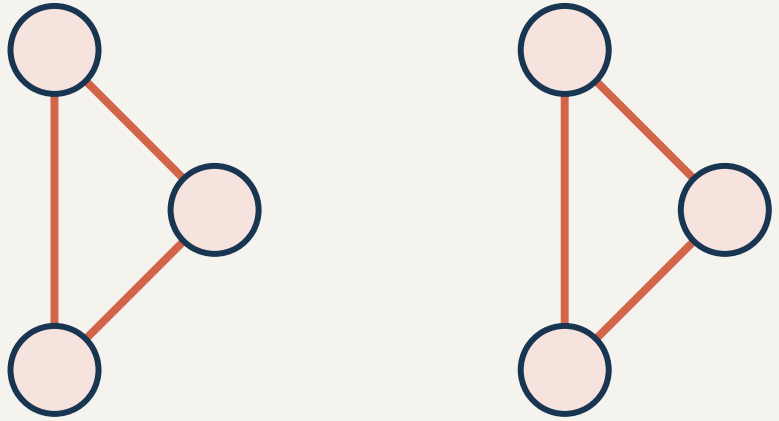
Local neighborhoods can hide global topology

UNIFORM SIX-CYCLE



Same degree and uniform labels

TWO DISJOINT TRIANGLES



Different component and cycle counts

Question: do persistence and cycle features change graph classification?

Two molecular benchmarks

MUTAG is primary; PTC-MR provides supplementary evidence

Dataset	Graphs / classes	Class counts (−1, +1)	Nodes / edges	Atom / bond label types	Average nodes / edges
MUTAG	188 / 2	63, 125	3,371 / 3,721	7 / 4	17.93 / 19.79
PTC-MR	344 / 2	192, 152	4,915 / 5,054	18 / 4	14.29 / 14.69

WHAT THE GRAPHS ENCODE

- Atoms are nodes; bonds are edges
- Both datasets use categorical node and edge labels
- Methods use node labels and topology; bond labels are ignored

TASK CONTEXT

MUTAG: mutagenicity of aromatic nitro compounds

PTC-MR: carcinogenicity classification for male rats

Dataset statistics recomputed from the local original TU archives and checked against _data.json.

Two papers define the method progression

Neighborhood refinement first; persistent topology next

2011 / SHERVASHIDZE ET AL.

Weisfeiler–Lehman Graph Kernels

Efficient graph features from iterative neighborhood-label refinement and subtree counts.

Contribution: scalable explicit kernel features

2019 / RIECK, BOCK & BORGWARDT

Persistent Weisfeiler–Lehman

Adds filtration-based component persistence and cycle contributions to WL features.

Contribution: persistent H_0 and H_1 information



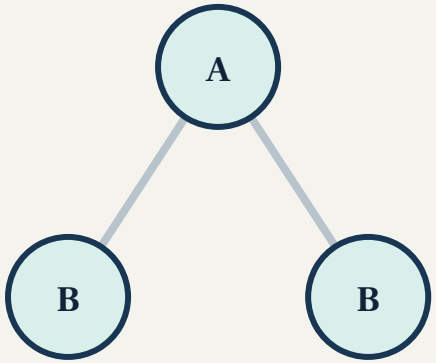
WL neighborhood counts → persistent components ✦ cycles

P-WL-C is established literature. λ weighting and binary cycle encoding are project adaptations.

Weisfeiler–Lehman refines node labels

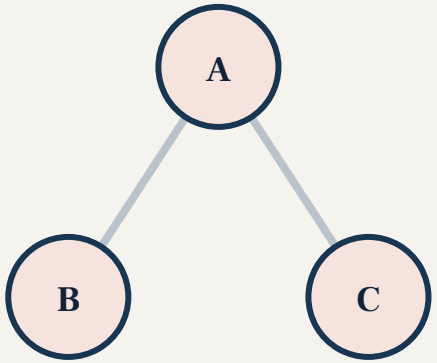
A node signature records its label and the multiset around it

NEIGHBORHOOD A



A | {B, B}

NEIGHBORHOOD B



A | {B, C}

$$\ell_h(v) = \text{hash}(\ell_{h-1}(v), \text{sorted neighbor labels})$$

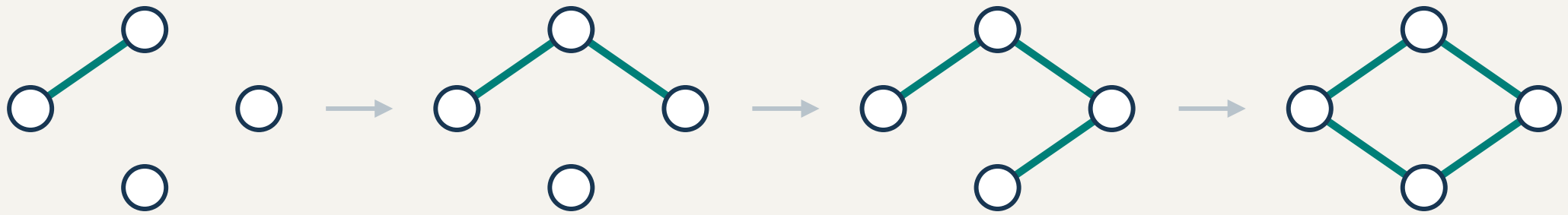
Repeat for h = 1…H

φWL(G) concatenates label counts from every depth.

Explicit graph vector → inner product is the induced WL kernel

Persistent WL weights label occurrences by merge lifetime

A filtration inserts edges from low weight to high weight



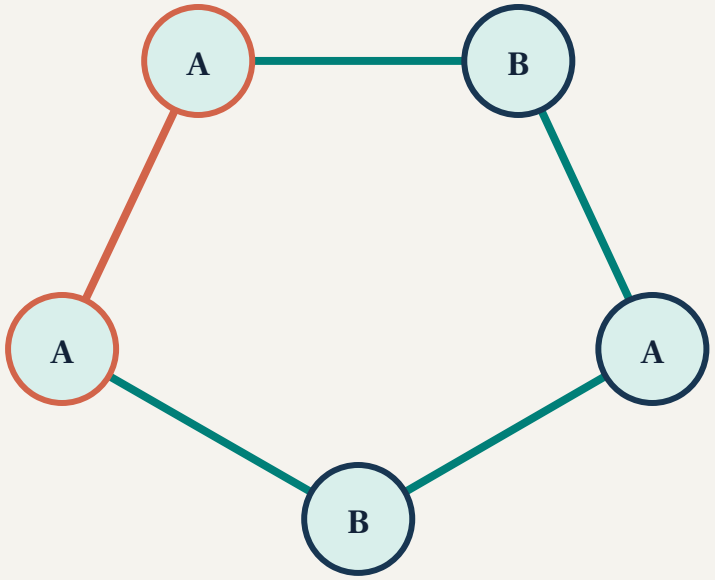
$$V_h(a) = \Sigma \text{ persistence}(v)^p \quad \text{for vertices with refined label } a$$

$p \in \{1, 2\}$ · concatenate across depths 0…H · p controls edge distance and contribution power

Unlike raw counts, a label's feature value also reflects how long its vertices remain connected.

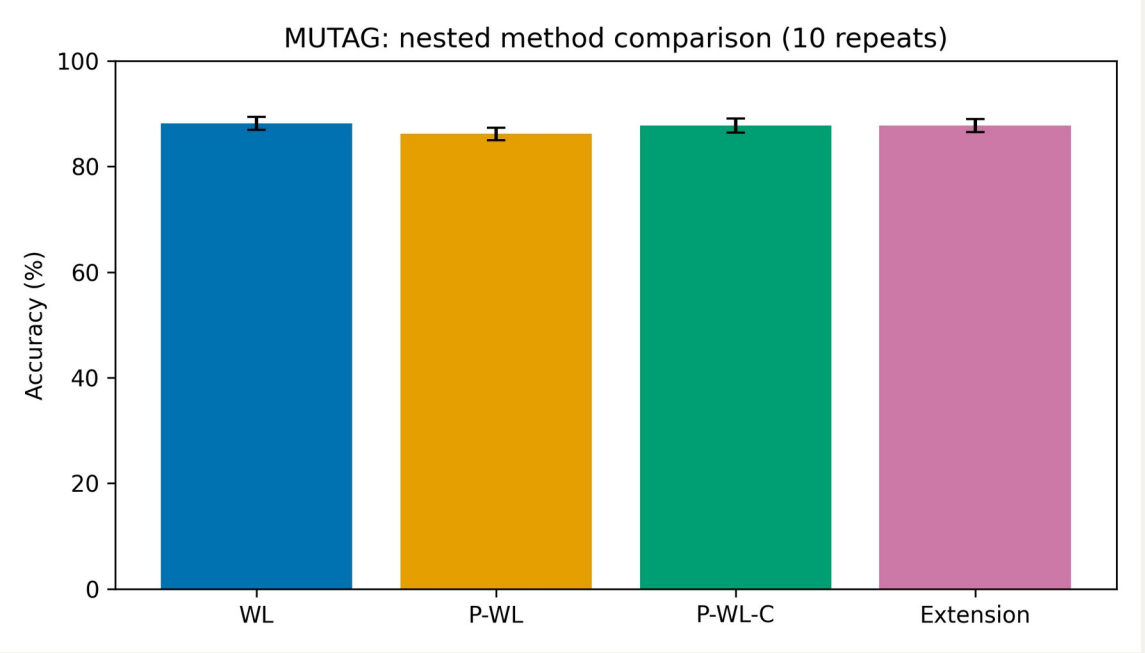
Cycle contributions add a second topological signal

A cycle appears when a new edge joins vertices already connected



CLOSING EDGE CREATES H_1

$C_h(a)$ adds the closing edge weight to its endpoint-label bins.



MUTAG ✦1.61 pp vs P-WL

PTC-MR · 0.55 pp vs P-WL

Same-label endpoints contribute twice. No triangles are filled.

Tested positive λ values matched predictions

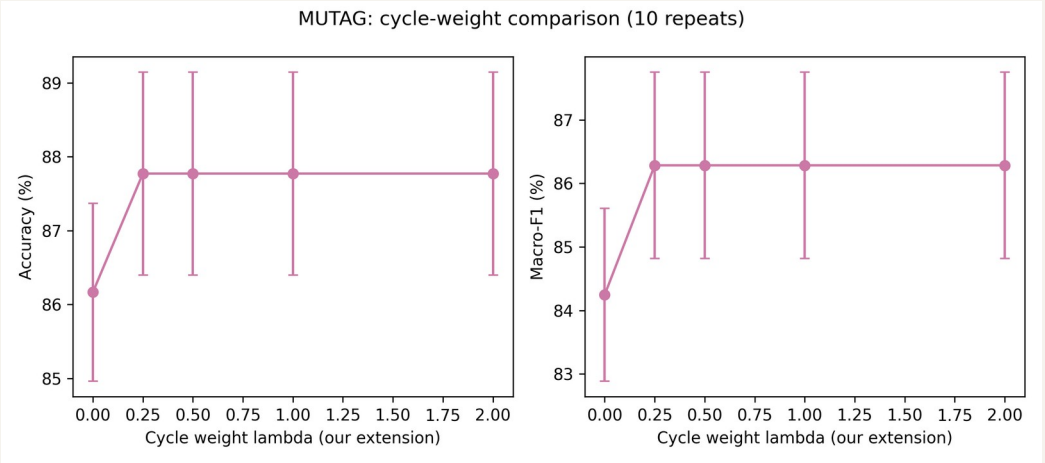
Observed across 100 outer folds per dataset; no universal guarantee

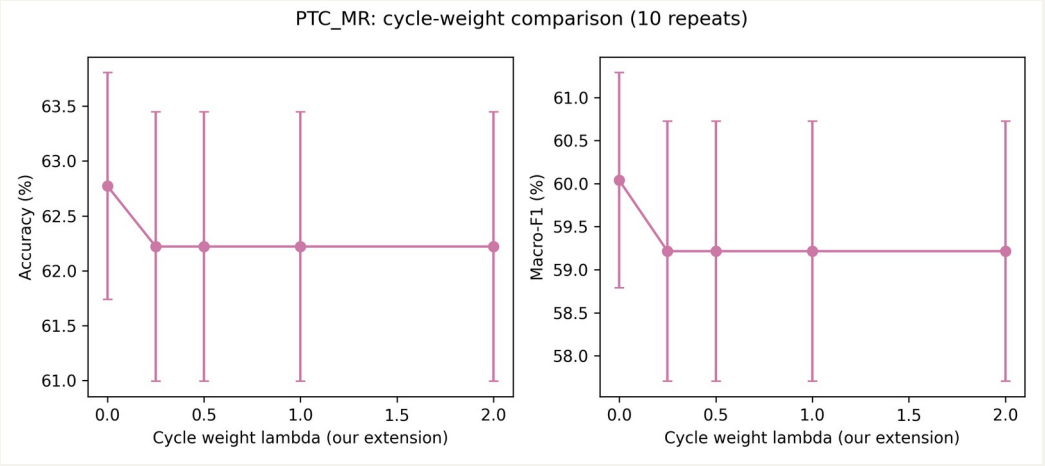
REPRESENTATION

$$\varphi\lambda = [\, V\, ,\, \lambda C\,]$$

$\lambda = 0$	P-WL
$\lambda = 0.25$	P-WL-C
$\lambda = 0.5$	P-WL-C
$\lambda = 1$	P-WL-C
$\lambda = 2$	P-WL-C

Identical predictions across 100 tested folds per dataset





Binary cycle encoding asks whether magnitude matters

The transformation keeps cycle presence and collapses positive magnitudes

BEFORE / MAGNITUDE

Cycle block C

0	2	8	0	3

B = 1[C≠0]

AFTER / PRESENCE

Binary cycle block B

0	1	1	0	1

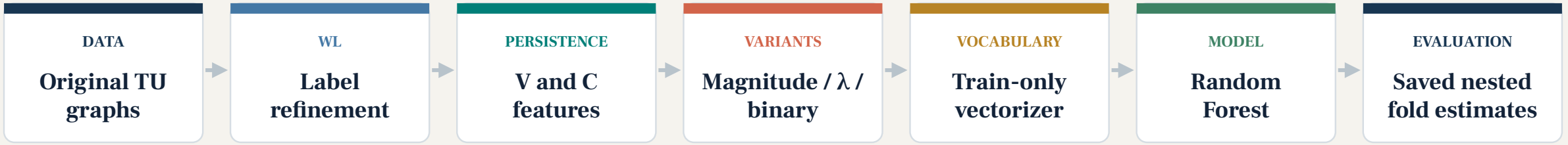
The persistence block V stays unchanged; only the completed cycle coordinates are binarized.

Magnitude → presence

MUTAG accuracy: −1.72 pp

One cached feature pipeline supports all variants

Feature vocabulary is learned inside each training split



DATA.PY

TU archive loading

WL.PY

Refinement and counts

PWL.PY

Filtration, V and C

EVALUATE.PY

Splits, RF and checkpoints

BINARY_CYCLE.PY

MUTAG cache-only extension

FINALIZE_CASE_STUDY.PY

Prediction audit and figures

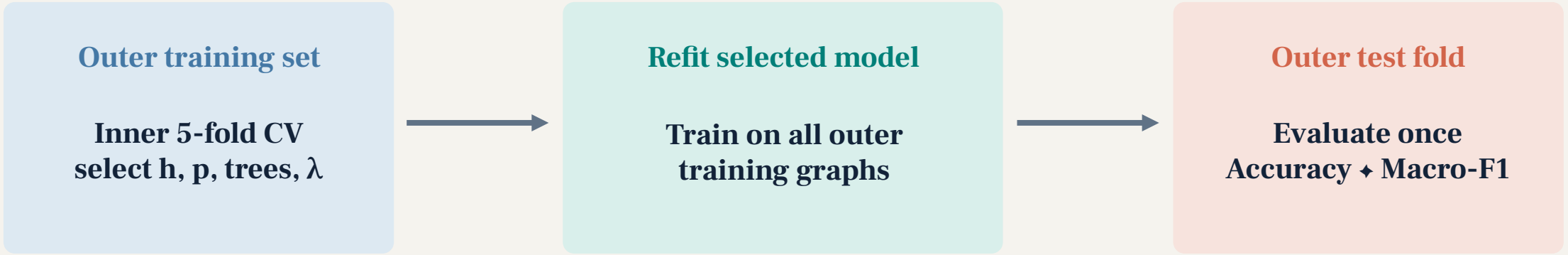
No global feature matrix: DictVectorizer refits on each inner and outer training split.

Nested cross-validation separates selection from testing

Each outer test fold stays untouched during model selection

10 repeats × 10 outer folds = 100 held-out evaluations per dataset

ONE OUTER SPLIT



SEARCH

h = 0…6 · p = 1,2 · 25/50/100 trees

FOREST

Gini · sqrt features · bootstrap · seed 42

SELECTION

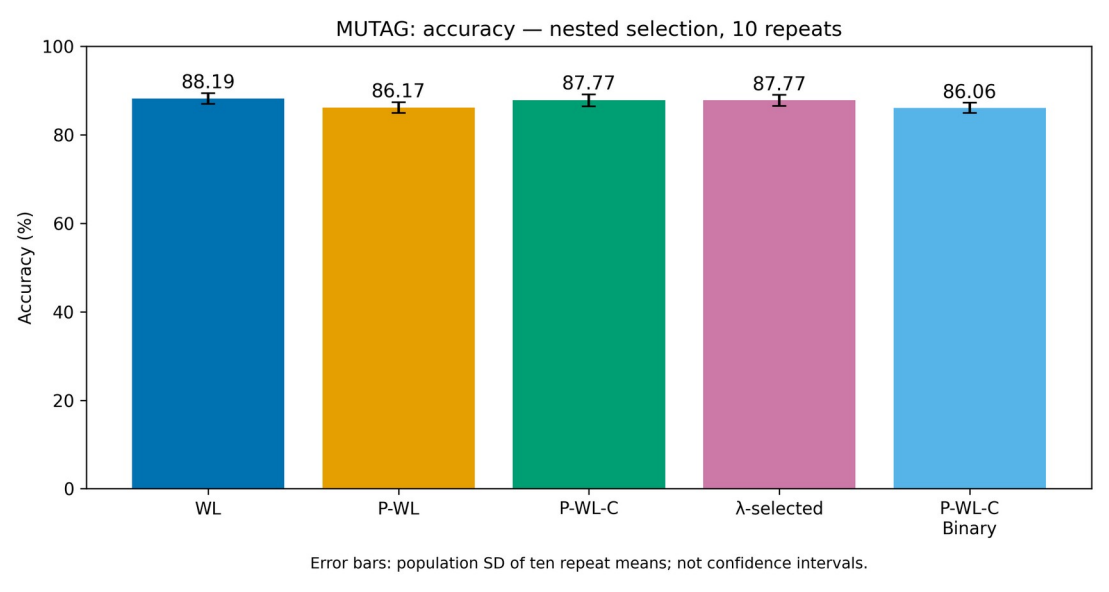
Mean inner accuracy; deterministic tie order

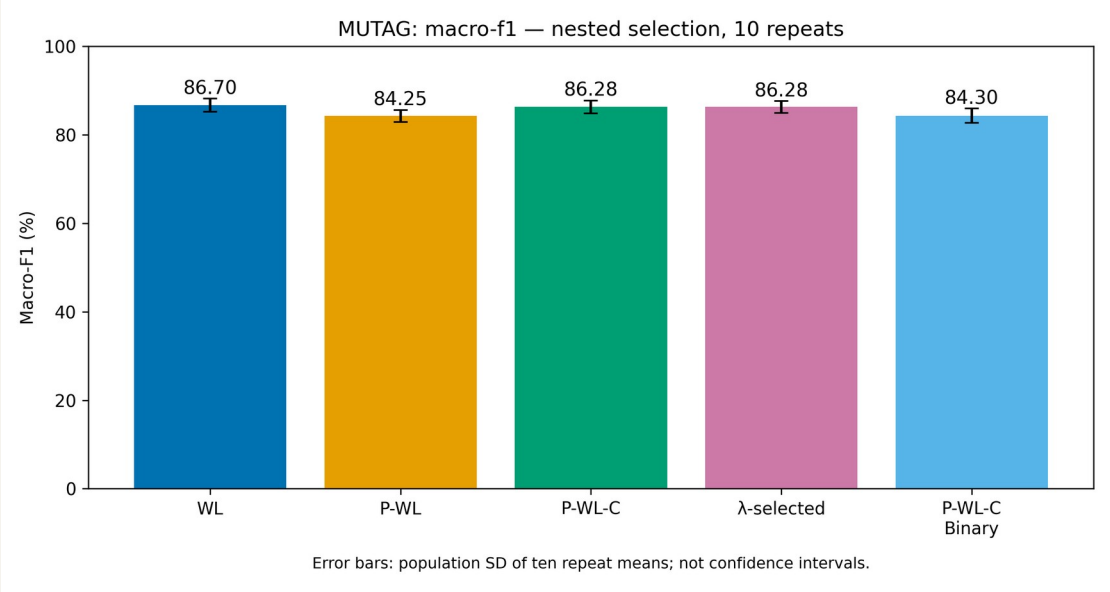
Reported variability is population SD over ten fold-averaged repeat means; it is descriptive.

MUTAG: WL leads; cycles recover part of the persistence gap

Mean ± SD across ten repeat-level estimates, in percent

Method	Accuracy (%)	Macro-F1 (%)
WL	88.19 ± 1.23	86.70 ± 1.46
P-WL	86.17 ± 1.20	84.25 ± 1.36
P-WL-C	87.77 ± 1.37	86.28 ± 1.47
λ-selected	87.77 ± 1.26	86.28 ± 1.35
P-WL-C-Binary	86.06 ± 1.16	84.30 ± 1.60

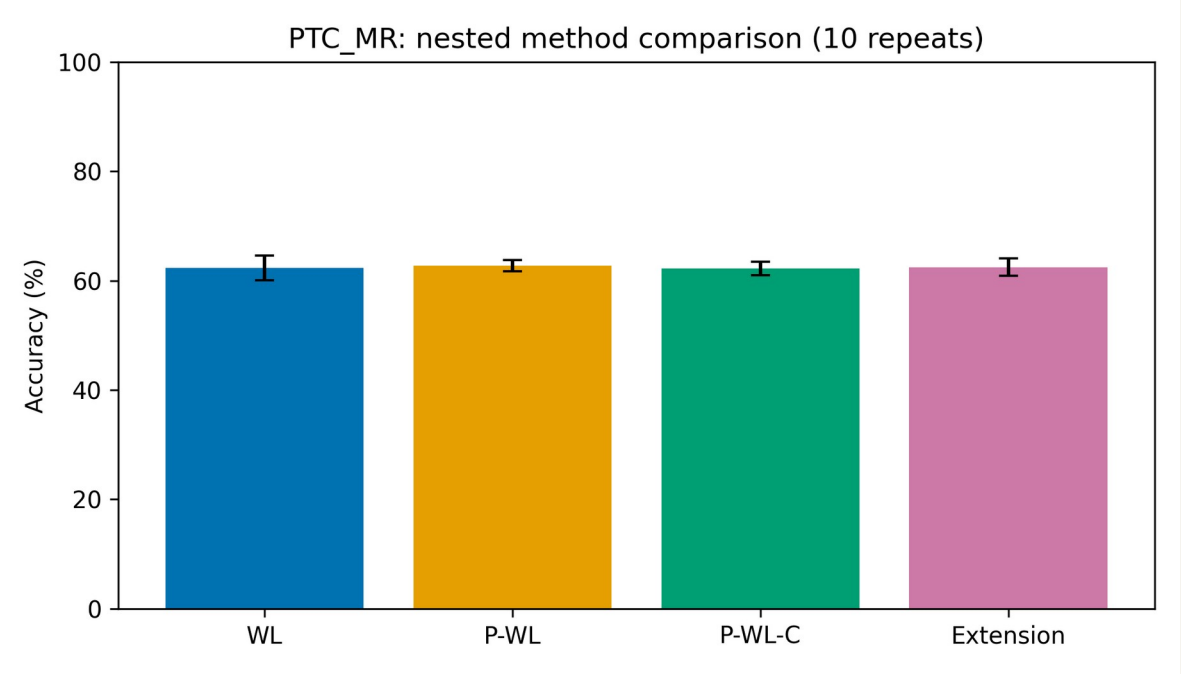




PTC-MR: cycles did not improve the selected model

Binary was evaluated only on primary MUTAG; PTC-MR provides supplementary baseline comparisons

Method	Accuracy (%)	Macro-F1 (%)
WL	62.33 ± 2.25	59.54 ± 2.47
P-WL	62.77 ± 1.03	60.04 ± 1.25
P-WL-C	62.22 ± 1.23	59.21 ± 1.51
λ-selected	62.49 ± 1.55	59.62 ± 1.87
P-WL-C-Binary	**Not evaluated**	**Not evaluated**

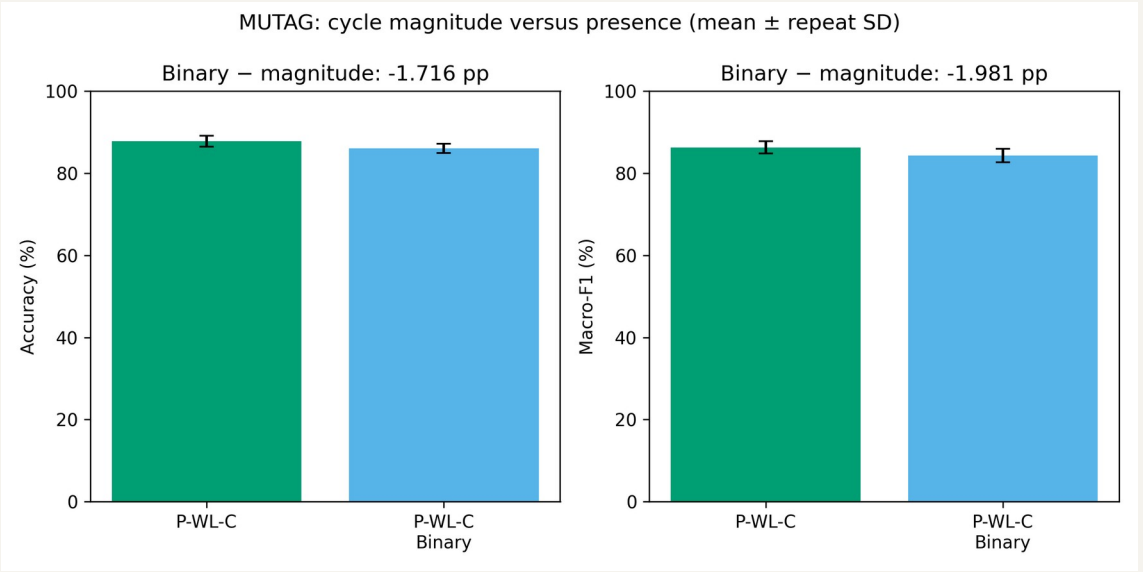


Binary cycle encoding: NOT RUN · OUT OF SCOPE

Each change altered the representation; effects varied by dataset

Paired score differences are descriptive comparisons of selected pipelines

MUTAG change	Accuracy Δ (pp)	Macro-F1 Δ (pp)
WL → P-WL	−2.02632	−2.45825
P-WL → P-WL-C	+1.60526	+2.03762
P-WL-C → λ-selected	−0.00292	−0.00751
P-WL-C → Binary	−1.71637	−1.98133



PTC-MR SUPPLEMENTARY

Change	Accuracy	Macro-F1
WL → P-WL	+0.44034	+0.49549
P-WL → P-WL-C	−0.55210	−0.82427
P-WL-C → λ-selected	+0.26471	+0.40320

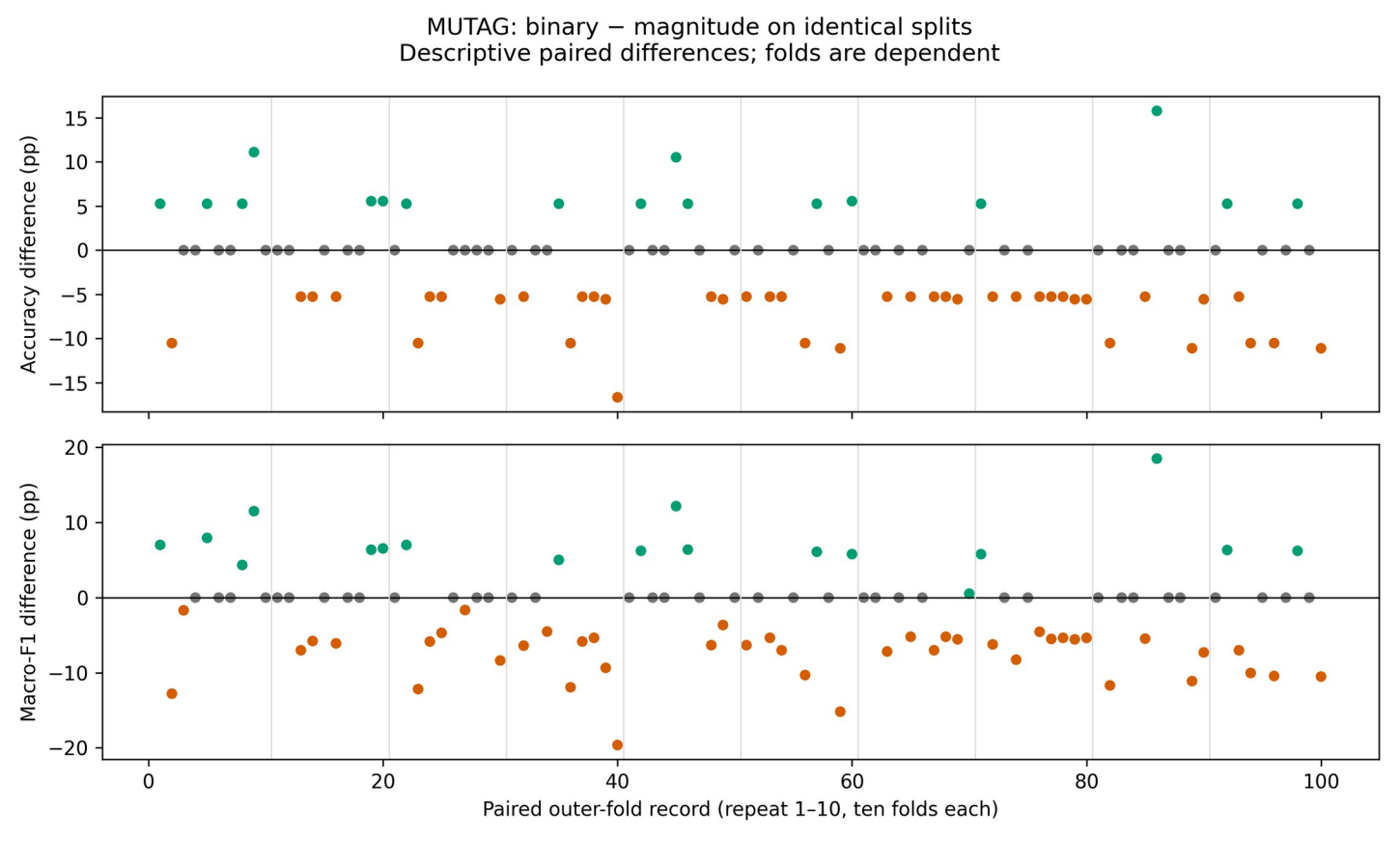
OBSERVED EFFECT

Cycles helped over P-WL on MUTAG and hurt on PTC-MR. Positive λ scaling stayed neutral. Binary fell 1.72 pp in MUTAG accuracy.

Differences compare independently selected nested pipelines; they are descriptive, not significance tests.

Binary encoding changed predictions across many folds

Most paired folds were mixed or tied; the average change was negative



BINARY VS P-WL-C

67 / 100

folds with changed predictions

SELECTED SETTINGS CHANGED

96 / 100 folds

h: 83 · p: 47 · trees: 67

114 of 1,880 graph-prediction opportunities changed; accuracy paired outcomes: 17 up, 41 down, 42 tied.

The study meets implement–modify–measure with a mixed result

Negative results are valid research findings; binary encoding reduced accuracy

Established methods

WL, P-WL and P-WL-C implemented from their papers. Cycle features are a paper method.

Tested adaptations

λ scaling had no effect at positive values. Binary encoding reduced MUTAG accuracy.

Measured evidence

Cycles helped over P-WL on MUTAG, hurt on PTC-MR, and stayed below WL on MUTAG.

P-WL-C-Binary: 86.06 ± 1.16% accuracy · 84.30 ± 1.60% Macro-F1

Limits: two small datasets, dependent repeated folds, descriptive SD, no significance test, and binary evaluated only on MUTAG.

Future work: independent benchmarks, edge-label-aware features, and other classifiers.

References and local evidence

Method definitions, dataset origins and saved evidence

1 Shervashidze, N. et al. (2011). “Weisfeiler-Lehman Graph Kernels.” Journal of Machine Learning Research, 12(77), 2539–2561.

https://www.jmlr.org/papers/v12/shervashidze11a.html

2 Rieck, B., Bock, C. & Borgwardt, K. (2019). “A Persistent Weisfeiler-Lehman Procedure for Graph Classification.” ICML, PMLR 97, 5448–5458.

https://proceedings.mlr.press/v97/rieck19a.html

3 BorgwardtLab/P-WL. Authors’ implementation; snapshot and source hashes retained locally.

https://github.com/BorgwardtLab/P-WL

4 TUDataset catalogue and bibliography. Dataset origins: Debnath et al. (1991), J. Med. Chem. 34(2), 786–797; Helma et al. (2001), Bioinformatics 17, 107–108.

https://chrsmrrs.github.io/datasets/docs/datasets/

Experiment claims trace to saved local folds and repeat means, not to published paper accuracies.